

2 (a) Gravitational force on planet m = $F_G = \frac{GMm}{x^2}$

The gravitational force is used as centripetal force by planet. $F_c = m\omega^2 x$

$$F_G = F_c$$

$$\frac{GMm}{x^2} = m\omega^2 x$$

$$GM = \omega x^3$$

(b) (i) 49.0 light years = speed of light x 3600 x 24 x 365 x 49
 = 4.636×10^{17} m = 4.64×10^{14} km

(ii) $\omega^2 x^3 = GM$

$$4\pi^2 x^3 = T^2 GM, \text{ hence } x^3 \propto T^2$$

the nearest planet to the star will have the shortest period.

nearest to $\mu - Arae$ $\mu - Arae$ c

$\mu - Arae$ d

$\mu - Arae$ b

furthest from $\mu - Arae$ $\mu - Arae$ e

(iii) $\mu - Arae$ c $[\omega^2 x^3]_c = GM$ -----(1)

$\mu - Arae$ e $[\omega^2 x^3]_e = GM$ -----(2)

from (1),

$$\left(\frac{2\pi}{9.69 \times 24 \times 60 \times 60} \right)^2 (1.35 \times 10^{10})^3 = GM$$

$$M = 2.078 \times 10^{30} \text{ kg}$$

from (2),

$$\left(\frac{2\pi}{4210 \times 24 \times 60 \times 60} \right)^2 (7.83 \times 10^{11})^3 = GM$$

$$M = 2.147 \times 10^{30} \text{ kg}$$

$$\text{Average } M = 2.11 \times 10^{30} \text{ kg}$$

3 (a) from the graph, the amplitude of oscillation, x_0 , is 2.4 cm or 0.024 m, maximum KE is